3AL Tutorial 1 2024

1. (a) Prove that $Z(D_4) = \{1, r^2\}$.

   (b) Determine the order of each element in D_4.

2. (a) Draw the symmetries of an equilateral triangle. (Something similar to the diagram in Example 1.7 of the notes)

   (b) Find a group isomorphic to D_3.

   Note: You don't have to write a full proof that the two groups are isomorphic. Just write a convincing statement of why a certain group is isomorphic to D_3.

3. Let $\varphi : G \to G'$ be a homomorphism. Prove the following:

   (a) $\ker \varphi$ is a subgroup of G,

   (b) φ is a monomorphism if and only if $\ker \varphi = \{1_G\}$,

   (c) $\operatorname{im} \varphi$ is a subgroup of G',

   (d) φ is an epimorphism if and only if $\operatorname{im} \varphi = G'$.

4. Let G be a cyclic group. Consider the map $\varphi : G \to G$ defined by $\varphi(a) = a^2$ for all $a \in G$.

   (a) Prove that φ is a homomorphism.

   (b) Determine the possible orders of elements in $\ker \varphi$.

5. Given the following groups and subgroups, list all of the distinct left cosets of the given subgroups (include a list of all elements in each coset).

   (a) $G = \mathbb{Z}_{21}$ with $H = \langle [3] \rangle$,

   (b) $G = \langle a \rangle$, where $|a| = 20$, with $H = \langle a^5 \rangle$.

6. Consider the (additive) group $\mathbb{Q}$ with subgroup $\mathbb{Z}$.

   (a) Show that all cosets of $\mathbb{Z}$ of the following form are distinct, where $n \in \mathbb{N}^+$,

$$\frac{1}{n} + \mathbb{Z} = \left\{ \frac{1}{n} + z : z \in \mathbb{Z} \right\}.$$

NOTE: Distinct means if $n \neq m$, then $\frac{1}{n} + \mathbb{Z} \neq \frac{1}{m} + \mathbb{Z}$.

(b) Prove or disprove: $|\mathbb{Q} : \mathbb{Z}|$ is finite.